

Computational modeling for cardiovascular tissue engineering: the importance of including cell behavior in growth and remodeling algorithms

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Abstract

Understanding cardiovascular growth and remodeling (G&R) is fundamental for designing robust cardiovascular tissue engineering strategies, which enable synthetic or biological scaffolds to transform into healthy living tissues after implantation. Computational modeling, particularly when integrated with experimental research, is key for advancing our understanding, predicting the *in vivo* evolution of engineered tissues, and efficiently optimizing scaffold designs. As cells are ultimately the drivers of G&R and known to change their behavior in response to mechanical cues, increasing efforts are currently undertaken to capture (mechano-mediated) cell behavior in computational models. In this selective review, we highlight some recent examples that are relevant in the context of cardiovascular tissue engineering and discuss the current and future biological and computational challenges for modeling cell-mediated G&R.

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Keywords

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Introduction

The ultimate goal of (*in situ*) cardiovascular tissue engineering (CVTE) is to regenerate cardiovascular tissues to restore or improve tissue function [1–3]. Current main approaches involve the use of a (synthetic)

biodegradable scaffold, which acts as temporary support for infiltrating cells that create their own, ideally native-like extracellular matrix (ECM) environment while the scaffold degrades over time. Such living engineered tissues can potentially overcome the limitations of current cardiovascular replacements, owing to their intrinsic ability to grow (i.e. change in mass and volume) and remodel (i.e. change in material properties) in response to changes in demands. However, current tissue engineering approaches still suffer from some limitations. A mechanistic understanding of the processes mediating functional and pathological growth and remodeling (G&R) is therefore conditional to design novel scaffolds that can guide these processes towards physiological regeneration and preserved long-term functionality.

Mechanical factors are increasingly recognized as important drivers of G&R [4,5]. Computational models capturing the mechanics and mechanobiology of cardiovascular tissues are therefore indispensable to mechanistically understand and predict the complex interplay between mechanics and cardiovascular G&R. In fact, recent studies have clearly demonstrated that integrating computational modeling into the experimental workflow leads to a substantially improved understanding and superior preclinical outcomes of novel tissue engineering approaches [6–9].

Extensive research in the computational biomechanics and mechanobiology field has been devoted to devising constitutive laws describing the biomechanical behavior of cardiovascular ECM (for some recent reviews, see Refs. [10–12]), and to developing algorithms that capture mechanically driven G&R of ECM components (e.g. see Refs. [8,13–16]). These developments are essential for accurately estimating overall tissue functionality and the mechanical cues responsible for tissue G&R. Concurrently, given that cells are key drivers of G&R and a conditional factor discriminating living from nonliving materials, increasing efforts are currently undertaken to model various types of cell behavior as well. This is particularly important for CVTE, as an increased understanding of cellular processes and their downstream effects on ECM G&R may reveal opportunities for steering cell behavior and consequently ECM G&R via rational adaptations in the scaffold design.

In this selective review, we highlight some examples of the current efforts and future challenges in modeling cell behavior that are relevant in the context of CVTE. Particularly, we focus on the aspects of (1) cytoskeletal remodeling, (2) cell turnover and migration, and (3) cell signaling, followed by a more general discussion on the future biological and computational challenges regarding modeling cell-mediated G&R.

Cytoskeletal remodeling

Cellular contractility has evident effects on tissue remodeling, as cellular forces can, for example, lead to tissue compaction (Figure 1A) and reorientation of collagen fibers (Figure 1B) [17,18]. Stress fibers are the main contractile component of the cytoskeleton of nonmuscle cells, such as heart valve interstitial cells [19]. Consequently, the orientation and maturation of these actomyosin bundles, respectively, determine the direction and magnitude of cellular forces (Figure 1B). However, modeling and predicting stress fiber organization is very complex because it remodels in response to several topographical and mechanical stimuli ([20] for a review), respectively provided, for example, by collagen fibers and pulsatile blood flow. For a better understanding, increasing efforts have been drawn toward developing computational models for stress fiber remodeling [21–25].

Recently, building on previous models [21,23], our group has developed numerical algorithms to predict stress fiber orientation in response to both topographical and mechanical stimuli [24,26]. By coupling these models with equations describing collagen turnover and prestretch as a result of stress fiber contraction, we could simulate the remodeling of native and tissue-engineered heart valves. Simulations of native heart valves suggested that cellular forces and alignment in response to mechanical stimuli are crucial for the emergence of the physiological collagen alignment at fetal age, whereas cellular alignment in response to topographical stimuli provided by collagen fibers is fundamental to maintain and reinforce the physiological collagen alignment during infancy [27]. Simulations of tissue-engineered heart valves were adopted to elucidate the role of cell contractility in the retraction of tissue-engineered leaflets shortly after implantation and to provide guidelines to avoid this unwanted phenomenon [28]. Further simulations were central to guide and improve the remodeling of tissue-engineered heart valves, such that they can now maintain excellent functionality up to one year after implantation [7]. Overall, these studies highlight the potential of analyzing stress fiber remodeling to achieve an increased understanding of cardiovascular tissue remodeling and eventually improve remodeling outcomes.

The computational models that we have developed to predict stress fiber remodeling [24,26] are mainly based on previous *in vitro* observations and only partly motivated by bio-chemo-mechanical considerations, which might limit their predictive potential in unexpected scenarios. More recently, by modifying a previous thermodynamically motivated framework for stress fiber remodeling [25], Chen et al. [29] proposed a new explanation for the orientation of stress fibers in response to mechanical stimuli. This novel thermodynamic model could be integrated with models for cell-mediated remodeling of cardiovascular tissues to provide a more physically motivated understanding, as already carried out for the investigation of tendon remodeling [30]. Furthermore, future studies could include the remodeling of additional (cytoskeletal) components that interact with stress fibers and contribute to regulate their organization and contractility in response to both topographical and mechanical cues, such as focal adhesions [31], microtubules [32], and vimentin [33]. Analyzing these interactions might inspire new approaches to engineer functional cardiovascular tissues and might elucidate pathologies associated with a dysregulation of these cellular components.

Cellular migration and proliferation

Cellular migration and proliferation can lead to changes in tissue volume, composition, and organization. For example, restenosis of arterial stents and vein grafts is correlated with excessive migration and proliferation of vascular smooth muscle cells in the intima layer [34]. While migrating, cells also secrete matrix metalloproteinases and exert forces, thereby causing ECM remodeling [35,36]. Accounting for migration and proliferation is therefore crucial to understand G&R of tissue-engineered cardiovascular grafts, as cell infiltration and proliferation are the main processes occurring at early stages after implantation (Figure 2).

Cell migration and proliferation can be modeled with continuum or agent-based models, depending on the degree of accuracy and efficiency required. Following a continuum approach, the local cell density can be captured by adopting a single partial differential equation accounting for chemo-attractants (for migration) and the local nutrient concentration (for mitosis/apoptosis) [37,38]. In contrast, agent-based models require a set of rules/equations to describe the movement and proliferation of individually modeled cell agents. The continuum approach is advantageous to investigate relatively large cell populations because it allows for considering the local cell concentration by solving a very limited number of equations. Agent-based models are often adopted when a relatively small population of cells is analyzed and a higher degree of accuracy to describe cell migration and proliferation is desired. For example, given their

importance for restenosis of coronary stents and vein grafts, several agent-based models have been recently proposed to accurately describe these two phenomena in this area [39–44].

In the context of tissue engineering, modeling cell migration and proliferation increases our understanding of the processes causing heterogeneity in the distribution of cells and collagen in tissue-engineered constructs, as recently demonstrated by Soares and Sacks [38]. By coupling several partial differential equations, their model considered local and temporal variations in oxygen concentration depending on the tissue-engineered construct porosity, permeability, and deformation; cell chemotaxis toward higher oxygen concentrations; and cell proliferation and ECM deposition dependent on the local oxygen concentration. In agreement with previous experiments, this parsimonious model predicted that mechanical stimulation of tissue-engineered samples increases the oxygen convection, which in turn decreases the transversal heterogeneity of the distribution of oxygen, cells, and ECM observed for statically cultured engineered tissues. However, as pointed out by the same authors in a different study [45], the proposed model cannot explain the recently demonstrated existence of an optimum strain for cell-mediated ECM formation [46]. Although cell-independent phenomena might also play a role, such as the optimum level of strain shielding against degradation observed for collagen [47], this observation invokes additional research. For example, the model of Soares and Sacks [38] could be enriched and adopted to test the hypothesis that the phenomena observed by D'Amore et al. [46] are a consequence of a cell mechanoresponsive behavior in terms of proliferation and apoptosis. Altogether, these studies [38,46] exemplify how a strong interplay between experiments and simulations can lead to new hypotheses and a better understanding of the G&R of engineered tissues, as already demonstrated for other cardiovascular tissues (we refer the reader to a review by Holmes [48] for examples).

Signaling pathways

Cellular phenomena such as cellular forces, migration, and proliferation are regulated not only by local mechanical cues but also by cell–cell signals from neighboring cells (Figure 3). For example, endothelial cells in the arterial intima layer can react to local cues by expressing growth factors (e.g. platelet-derived growth factor (PDGF)) that, while diffusing, influence the behavior of cells in the media [49]. Therefore, including cell–cell signals in computational models is crucial to predict how local cues influence the global G&R of cardiovascular tissues and to guide cells toward a

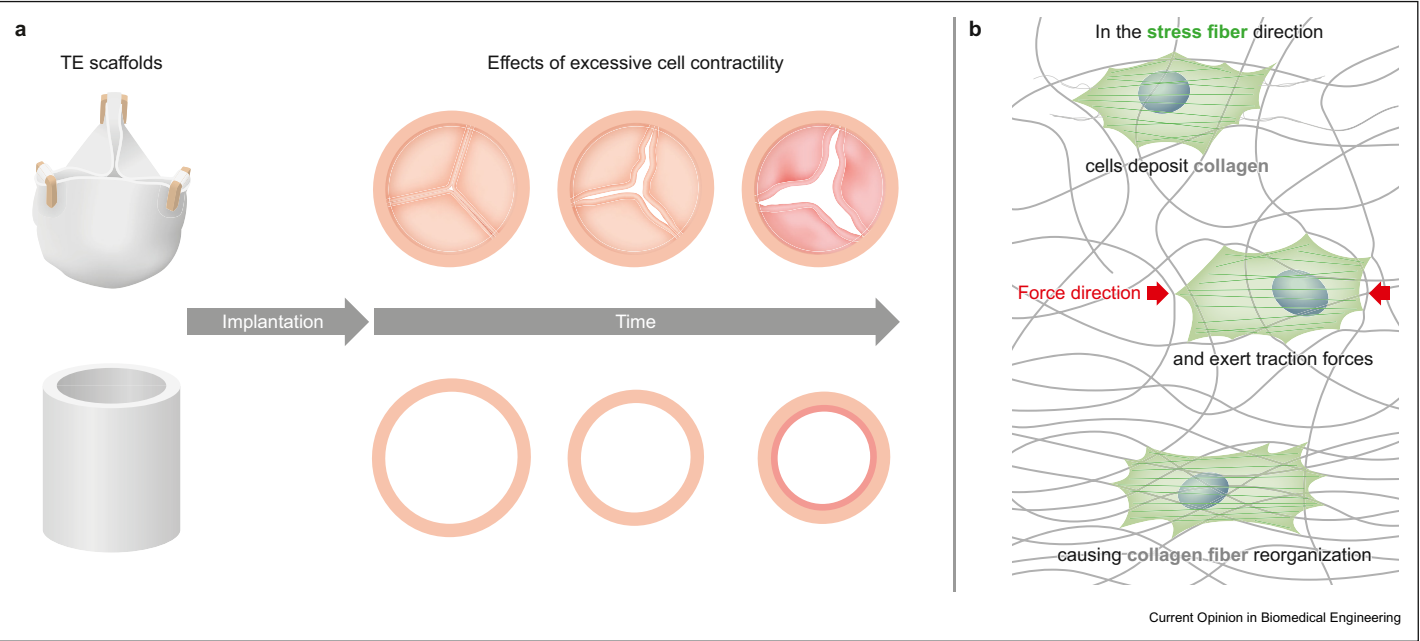
physiological G&R of tissue-engineered arteries (Figure 3A) and heart valves (Figure 3B).

Cell–cell signals can occur at different length scales and are as such classified as endocrine (long distances), paracrine (short distances), and juxtacrine signals (requiring cell–cell contact). In the context of G&R in the vasculature, paracrine and juxtacrine signaling are mostly analyzed (Figure 3C and D). On the one hand, paracrine signals can be modeled via continuum partial differential equations to account for the signal diffusion [50–53]. On the other hand, juxtacrine signals require the adoption of agent-based modeling [54], as this approach enables the consideration of individual cellular positions and thus the identification of the signaling network of contacting cells.

Models accounting for cell–cell signals can be adopted to investigate G&R as a result of communication among different cells. For example, Marino et al. [51,52] have recently proposed a computational framework for cell-mediated ECM remodeling in arteries accounting for the reaction-diffusion of paracrine signals. With this framework, the authors analyzed how the crosstalk between signals coming from macrophages in the adventitia, endothelial cells in the intima, and pharmacological treatments can influence the production of matrix metalloproteases by cells in the media, which in turn affects arterial remodeling. Given the importance of cell–cell signaling for the regulation of cell activity, a similar approach could be adopted to investigate and guide the remodeling of engineered tissues.

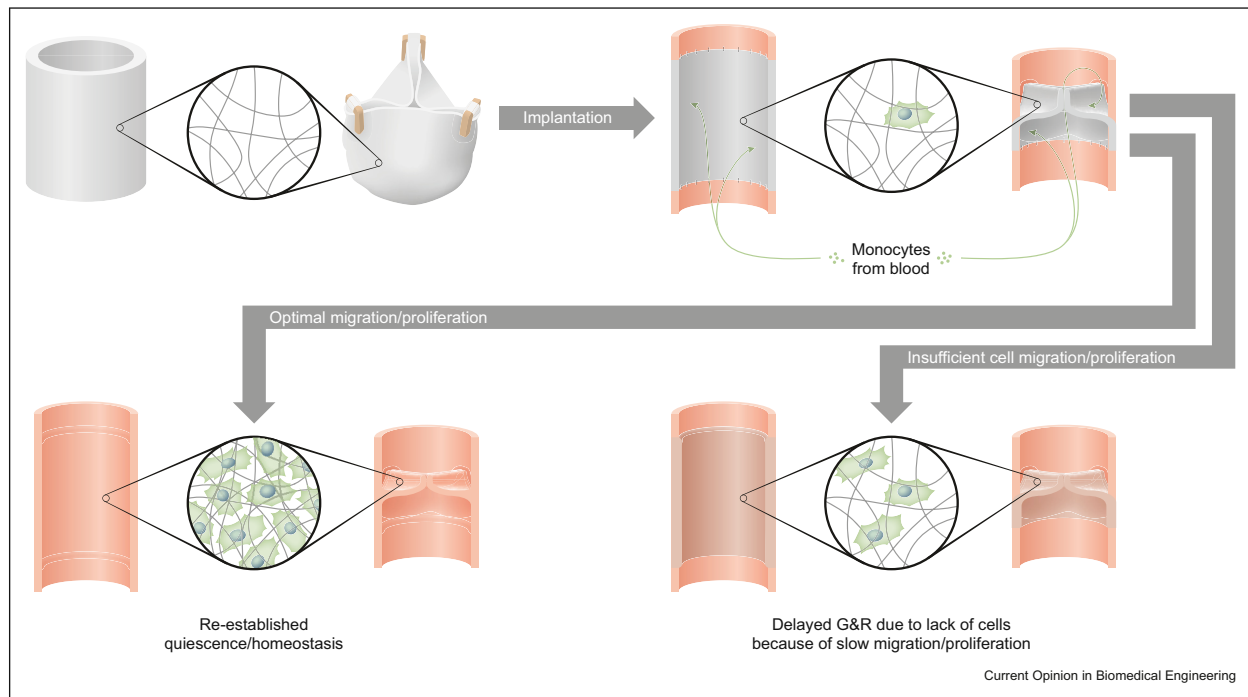
Cell activity is also influenced by juxtacrine signals. For example, transforming growth factor (TGF- β) seems to affect the regulation of the vascular smooth muscle cell phenotype via a crosstalk with Notch [55], a juxtacrine signal influencing several differentiation markers (Figure 3D). *In vitro* experiments have also shown that Notch is affected by cyclic strain [56]. As such, this signaling pathway is most likely crucial for cell-mediated G&R in response to mechanical stimuli. This highlights the importance of developing models for juxtacrine signaling in cardiovascular tissues, such as the model recently proposed by our group [54]. Interestingly, by accounting for the Notch mechanoresponse and the phenotypic modulation of Notch, this model suggested that the communication of arterial cells via Notch induces a collective phenotypic switch of cells from synthetic to contractile (and vice-versa) depending on the arterial wall thickness, which may be essential for the emergence of arterial homeostasis.

As the number of cross-talking signaling pathways that are known to regulate cell behavior will probably only



Effects of cell contractility on the evolution of engineered cardiovascular tissues. **(a)** Cellular contractility strongly influences the remodeling and functionality of tissue-engineered constructs upon im-plantation. Over time, excessive cellular contractility can cause compaction of tissue-engineered vascular grafts and heart valve leaflets. **(b)** Cellular contractility is strongly regulated by the remodeling of stress fibers: actomyosin bundles in the cellular cytoskeleton determine the direction of collagen fiber deposition and cellular forces, thereby strongly influencing collagen fiber (re)orientation.

Figure 2



Current *in situ* tissue engineering approaches rely on the potential of scaffolds to attract and favor the proliferation of native cells infiltrating from the recipient's circulation. An incorrect rate of migration and proliferation of host cells can lead to an unfavorable delay of the growth and remodeling process of engineered tissues.

increase over time, future efforts should also be directed toward identifying key signaling pathways and how exactly these are affected by mechanics. For example, signaling network models could be developed to identify the main drivers of specific cardiovascular cellular behavior, for example, as demonstrated by Zeigler et al. [57] for myofibroblasts. This will enable the reduction of the model rules and parameters, thereby decreasing the computational costs while increasing the confidence in the model predictions.

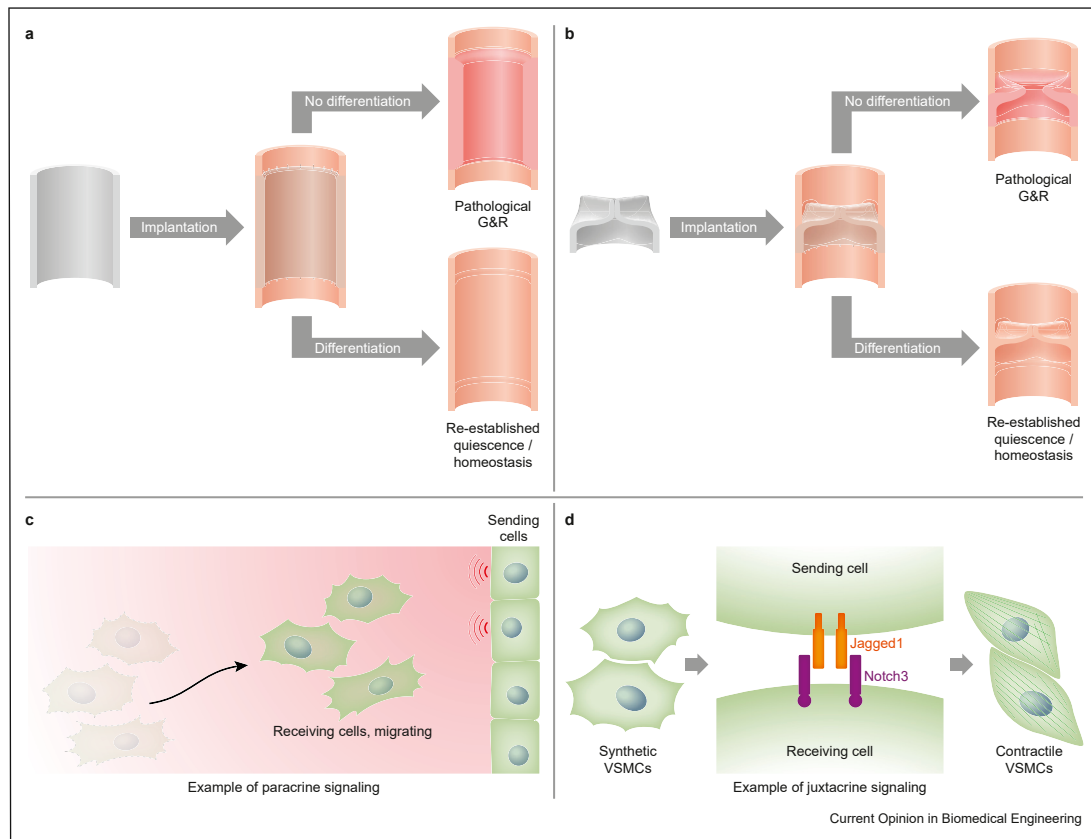
General challenges and future directions

The success of *in situ* CVTE primarily relies on the individual's regenerative capacity, which should enable the (synthetic) scaffold to transform into a living tissue with native-like form and function. Since this intrinsic regenerative capacity varies across individuals owing to subject-specific differences in age, gender, and potential comorbidities, the evolution of engineered grafts can substantially vary across individuals, with consequentially varying degrees of success [58,59]. Therefore, besides focusing on the development of computational models regarding the specific behavior of (healthy) ECM-producing cells, it is also important to model how cell-mediated G&R interacts with other regenerative processes as a function of patient-specific conditions.

The immune system is obviously a key player in regeneration, and *in vitro* [60,61] and *in vivo* [62,63] studies have clearly demonstrated that the inflammatory response during the early phases after scaffold implantation has a key influence on the ultimate G&R of engineered vascular grafts. Over the past years, the Humphrey lab made important progress in modeling the interplay between immuno-driven and mechano-mediated G&R. For example, Miller et al. [13,64] demonstrated that incorporating the transition from inflammatory-mediated tissue formation to mechano-biologically driven G&R is critical for correctly modeling the *in vivo* evolution of engineered vascular grafts. Recently, Szafron et al. [8] investigated the impact of variations in inflammatory response and showed that the moderate inflammatory response to the implanted construct in immuno-compromised mice is more favorable for long-term vessel development compared with the exuberant response exhibited by immuno-competent mice.

In the future, an increased focus on including specific immune cell behavior may lead to novel insights in this area and potentially to tools to direct the immune response toward functional regeneration. Several *in vitro* experiments have demonstrated that macrophages, key players in the foreign body response, are

Figure 3



Regulation and impact of cellular phenotypes. Proper regulation of the phenotype of cells infiltrating scaffolds is a key requirement to avoid excessive thickening of tissue-engineered arteries (a) and heart valve leaflets (b). Therefore, after implantation, a quiescent cellular phenotype should be induced when the transformation of the scaffold toward a native-like tissue has been completed. The regulation of cellular phenotype, as well as cellular migration and contractility, is strongly influenced by cell–cell communication. For example, paracrine signaling can influence cell migration (c), whereas juxtacrine signaling (e.g. via Notch) induces differentiation of synthetic vascular smooth muscle cells toward their contractile phenotype (d).

mechanosensitive and can change their phenotype depending on how they are mechanically loaded [65–67]. Including macrophage mechanosensitivity and other relevant cellular phenomena will be important for obtaining a mechanistic understanding of how immune cells, in conjunction with (recruited) tissue cells, drive the cell-mediated *in vivo* evolution of engineered grafts.

On the other hand, the inclusion of increasingly detailed cell behavior in computational models poses several challenges related to computational costs, parameter identification, and model validation. Efforts aimed at more efficiently solving the model equations or approximating their solutions [14,68–70] are essential to ensure that models remain numerically tractable when applied to simulating G&R under complex *in vivo* conditions. The challenges related to parameter identification and model validation can be partially resolved when more, systematic, and quantitative experimental data become available. Still, it is important to minimize the number of mechanisms and model parameters

depending on the research question under investigation to balance model complexity and uncertainty, and consequently maximize model credibility [71].

Summary and outlook

Understanding cardiovascular G&R is fundamental for designing successful and robust *in situ* CVTE strategies. Notwithstanding the essential role of experimental research, computational modeling is key for advancing our understanding and particularly our predictive capability of G&R. As cells are the ultimate drivers of G&R, increasing efforts are currently undertaken to model various types of mechano-mediated cell behavior. In this selective review, based on their applicability to the CVTE field, we highlighted some relevant examples of recent developments in modeling cytoskeletal remodeling, cell turnover and migration, and cell signaling and discussed some of the current and future challenges for advancing the computational mechanobiology field.

When combined with algorithms that describe G&R of the ECM, these cell-mediated G&R models will enable further optimization of scaffold properties [72,73], to ideally control cell behavior and optimize downstream matrix formation. Furthermore, these models will be crucial to estimate the sensitivity of CVTE outcomes to variations in patient-specific factors. To facilitate this process, it is important that the computational modeling field joins forces with experts in tissue engineering, cell biology, immunology, medicine, and materials science, as the multidisciplinary nature of tissue engineering [1] naturally calls for multidisciplinary research collaborations to advance the field.

Conflict of interest statement

Nothing declared.

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References

Papers of particular interest, published within the period of review, have been highlighted as:

- * of special interest
- ** of outstanding interest

1. Langer R, Vacanti JP: **Tissue engineering**. *Science* 1993, **260**: 920–926.
2. Best C, Onwuka E, Pepper V, Sams M, Breuer J, Breuer C: **Cardiovascular tissue engineering: preclinical validation to bedside application**. *Physiology* 2016, **31**:7–15.
3. Fernández-Avilés F, Sanz-Ruiz R, Climent AM, Badimon L, Bolli R, Charron D, Fuster V, Janssens S, Kastrup J, Kim H-S, et al.: **Global position paper on cardiovascular regenerative medicine**. *Eur Heart J* 2017, **38**:2532–2546.
4. Humphrey JD: **Vascular adaptation and mechanical homeostasis at tissue, cellular, and sub-cellular levels**. *Cell Biochem Biophys* 2008, **50**:53–78.
5. Humphrey JD, Dufresne ER, Schwartz MA: **Mechano-transduction and extracellular matrix homeostasis**. *Nat Rev Mol Cell Biol* 2014, **15**:802–812.
6. Khosravi R, Miller KS, Best CA, Shih YC, Lee Y-U, Yi T, Shinoka T, Breuer CK, Humphrey JD: **Biomechanical diversity despite mechanobiological stability in tissue engineered vascular grafts two years post-implantation**. *Tissue Eng A* 2015, **21**:1529–1538.
7. Emmert MY, Schmitt BA, Loerakker S, Sanders B, Priestersbach H, Fioretta ES, Bruder L, Brakmann K, Motta SE, Lintas V, et al.: **Computational modeling guides tissue-engineered heart valve design for long-term in vivo performance in a translational sheep model**. *Sci Transl Med* 2018, **10**, eaan4587.
8. Szafron JM, Khosravi R, Reinhardt J, Best CA, Bersi MR, Yi T, Breuer CK, Humphrey JD: **Immuno-driven and mechano-mediated neotissue formation in tissue engineered vascular grafts**. *Ann Biomed Eng* 2018, **46**:1938–1950.
9. Best CA, Szafron JM, Rocco KA, Zbinden J, Dean EW, Maxfield MW, Kurobe H, Tara S, Bagi PS, Udelman BV, et al.: **Differential outcomes of venous and arterial tissue engineered vascular grafts highlight the importance of coupling long-term implantation studies with computational modeling**. *Acta Biomater* 2019, **94**:183–194.
10. Holzapfel GA, Ogden RW: **Biomechanical relevance of the microstructure in artery walls with a focus on passive and active components**. *Am J Physiol Heart Circ Physiol* 2018, **315**: H540–H549.
11. Holzapfel GA, Ogden RW, Sherifova S: **On fibre dispersion modelling of soft biological tissues: a review**. *Proc R Soc A Math Phys Eng Sci* 2019, **475**:20180736.
12. Sacks MS, Drach A, Lee CH, Khalighi AH, Rego BV, Zhang W, Ayoub S, Yoganathan AP, Gorman RC, Gorman JH: **On the simulation of mitral valve function in health, disease, and treatment**. *J Biomech Eng* 2019. <https://doi.org/10.1115/1.4043552>.
13. Miller KS, Khosravi R, Breuer CK, Humphrey JD: **A hypothesis-driven parametric study of effects of polymeric scaffold properties on tissue engineered neovessel formation**. *Acta Biomater* 2015, **11**:283–294.
14. Cyron CJ, Aydin RC, Humphrey JD: **A homogenized constrained mixture (and mechanical analog) model for growth and remodeling of soft tissue**. *Biomechanics Model Mechanobiol* 2016, **15**:1389–1403.
15. Ambrosi D, Ben Amar M, Cyron CJ, DeSimone A, Goriely A, Humphrey JD, Kuhl E: **Growth and remodelling of living systems : perspectives, challenges and opportunities** 2019, **16**. 20190233.
16. Braeu FA, Aydin RC, Cyron CJ: **Anisotropic stiffness and tensional homeostasis induce a natural anisotropy of volumetric growth and remodeling in soft biological tissues**. *Biomechanics Model Mechanobiol* 2019, **18**:327–345.
17. Kim A, Lakshman N, Petroll WM: **Quantitative assessment of local collagen matrix remodeling in 3-D culture: the role of Rho kinase**. *Exp Cell Res* 2006, **312**:3683–3692.
18. Turlier H, Maître J-L: **Mechanics of tissue compaction**. *Semin Cell Dev Biol* 2015, **47–48**:110–117.
19. Sakamoto Y, Buchanan RM, Sacks MS: **On intrinsic stress fiber contractile forces in semilunar heart valve interstitial cells using a continuum mixture model**. *J Mech Behav Biomed Mater* 2016, **54**:244–258.
20. Sears C, Kaunas R: **The many ways adherent cells respond to applied stretch**. *J Biomech* 2016, **49**:1347–1354.
21. Deshpande VS, McMeeking RM, Evans AG: **A bio-chemo-mechanical model for cell contractility**. *Proc Natl Acad Sci U S A* 2006, **103**:14015–14020.
22. Kaunas R, Hsu HJ: **A kinematic model of stretch-induced stress fiber turnover and reorientation**. *J Theor Biol* 2009, **257**: 320–330.
23. Vernerey FJ, Farsad M: **A constrained mixture approach to mechano-sensing and force generation in contractile cells**. *J Mech Behav Biomed Mater* 2011, **4**:1683–1699.
24. Obbink-Huizer C, Oomens CWJ, Loerakker S, Foolen J, Bouten CVC, Baaijens FPT: **Computational model predicts cell**

- orientation in response to a range of mechanical stimuli. *Biomechanics Model Mechanobiol* 2014, **13**:227–236.
25. Vigliotti A, Ronan W, Baaijens FPT, Deshpande VS: **A thermodynamically motivated model for stress-fiber reorganization.** *Biomechanics Model Mechanobiol* 2016, **15**: 761–789.
 26. Ristori T, Notermans TMW, Foolen J, Kurniawan NA, Bouten CVC, Baaijens FPT, Loerakker S: **Modelling the combined effects of collagen and cyclic strain on cellular orientation in collagenous tissues.** *Sci Rep* 2018, **8**:8518.
 27. Ristori T, Bouten CVC, Baaijens FPT, Loerakker S: **Predicting and understanding collagen remodeling in human native heart valves during early development.** *Acta Biomater* 2018, **80**:203–216.
- A computational model for cell-mediated collagen remodeling unravels the mechanisms leading to the physiological collagen organization in native pediatric semilunar valves by accounting for stress fiber remodeling in response to topographical and mechanical cues.
28. Loerakker S, Ristori T, Baaijens FPT: **A computational analysis of cell-mediated compaction and collagen remodeling in tissue-engineered heart valves.** *J Mech Behav Biomed Mater* 2016, **58**:173–187.
 29. Chen K, Vigliotti A, Bacca M, McMeeking RM, Deshpande VS, Holmes JW: **Role of boundary conditions in determining cell alignment in response to stretch.** *Proc Natl Acad Sci U S A* 2018, **115**:986–991.
- Coupling between experiments and a thermodynamic model for stress fiber remodeling uncovers the role of boundary conditions for cell (re) orientation in response to mechanical cues.
30. Chen K, Hu X, Blemker SS, Holmes JW: **Multiscale computational model of Achilles tendon wound healing: untangling the effects of repair and loading.** *PLoS Comput Biol* 2018, **14**, e1006652.
 31. Ristori T, Vigliotti A, Baaijens FPT, Loerakker S, Deshpande VS: **Prediction of cell alignment on cyclically strained grooved substrates.** *Biophys J* 2016, **111**:2274–2285.
 32. Morioka M, Parameswaran H, Naruse K, Kondo M, Sokabe M, Hasegawa Y, Suki B, Ito S: **Microtubule dynamics regulate cyclic stretch-induced cell alignment in human airway smooth muscle cells.** *PLoS One* 2011, **6**, e26384.
 33. Jiu Y, Peränen J, Schaible N, Cheng F, Eriksson JE, Krishnan R, Lappalainen P: **Vimentin intermediate filaments control actin stress fiber assembly through GEF-H1 and RhoA.** *J Cell Sci* 2017, **130**:892–902.
 34. Clowes AW, Clowes MM: **Kinetics of cellular proliferation after arterial injury. IV. Heparin inhibits rat smooth muscle mitogenesis and migration.** *Circ Res* 1986, **58**:839–845.
 35. Newby AC: **Matrix metalloproteinases regulate migration, proliferation, and death of vascular smooth muscle cells by degrading matrix and non-matrix substrates.** *Cardiovasc Res* 2006, **69**:614–624.
 36. Provenzano PP, Inman DR, Eliceiri KW, Trier SM, Keely PJ: **Contact guidance mediated three-dimensional cell migration is regulated by Rho/ROCK-dependent matrix reorganization.** *Biophys J* 2008, **95**:5374–5384.
 37. Manzano S, Moreno-Loshuertos R, Doblaré M, Ochoa I, Hamdy Doweidar M: **Structural biology response of a collagen hydrogel synthetic extracellular matrix with embedded human fibroblast : computational and experimental analysis.** *Med Biol Eng Comput* 2015, **53**:721–735.
 38. Soares JS, Sacks MS: **A triphasic constrained mixture model of engineered tissue formation under in vitro dynamic mechanical conditioning.** *Biomechanics Model Mechanobiol* 2016, **15**:293–316.
 39. Garbey M, Rahman M, Berceli SA: **A multiscale computational framework to understand vascular adaptation.** *J Comput Sci* 2015, **8**:32–47.
 40. Tahir H, Niculescu I, Bona-Casas C, Merks RMH, Hoekstra AG: **An in silico study on the role of smooth muscle cell migration in neointimal formation after coronary stenting.** *J R Soc Interface* 2015, **12**:20150358.
 41. Garbey M, Casarin S, Berceli SA: **Vascular adaptation: pattern formation and cross validation between an agent based model and a dynamical system.** *J Theor Biol* 2017, **429**: 149–163.
 42. Zun PS, Anikina T, Svitenkov A, Hoekstra AG: **A comparison of fully-coupled 3D in-stent restenosis simulations to in-vivo data.** *Front Physiol* 2017, **8**:284.
 43. Nolan DR, Lally C: **An investigation of damage mechanisms in mechanobiological models of in-stent restenosis.** *J Comput Sci* 2018, **24**:132–142.
 44. Garbey M, Casarin S, Berceli SA: **A versatile hybrid agent-based, particle and partial differential equations method to analyze vascular adaptation.** *Biomechanics Model Mechanobiol* 2019, **18**:29–44.
- Development of a finite-element and agent-based model enabling to test hypotheses concerning the key role and cues of cellular migration in the context of intimal hyperplasia in vein grafts.
45. Soares JS, Zhang W, Sacks MS: **A mathematical model for the determination of forming tissue moduli in needled-nonwoven scaffolds.** *Acta Biomater* 2017, **51**:220–236.
 46. D'Amore A, Soares JS, Stella JA, Zhang W, Amoroso NJ, Mayer JE, Wagner WR, Sacks MS: **Large strain stimulation promotes extracellular matrix production and stiffness in an elastomeric scaffold model.** *J Mech Behav Biomed Mater* 2016, **62**:619–635.
 47. Huang C, Yannas IV: **Mechanochemical studies of enzymatic degradation of insoluble collagen fibers.** *J Biomed Mater Res* 1977, **11**:137–154.
 48. Holmes JW: **Model first and ask questions later: confessions of a reformed experimentalist.** *J Biomech Eng* 2019. <https://doi.org/10.1115/1.4043432>.
 49. Qi YX, Jiang J, Jiang XH, Wang XD, Ji SY, Han Y, Long DK, Shen BR, Yan ZQ, Chien S, et al.: **PDGF-BB and TGF- β 1 on cross-talk between endothelial and smooth muscle cells in vascular remodeling induced by low shear stress.** *Proc Natl Acad Sci U S A* 2011, **108**:1908–1913.
 50. Fok PW, Sanft R: **A biochemical and mechanical model of injury-induced intimal thickening.** *Math Med Biol* 2017, **34**: 77–108.
 51. Marino M, Pontrelli G, Vairo G, Wriggers P: **Coupling microscale transport and tissue mechanics: modeling strategies for arterial multiphysics.** In *Modeling of microscale transport in biological processes*. Edited by Becker S, Elsevier Inc.; 2017: 77–112.
 52. Marino M, Pontrelli G, Vairo G, Wriggers P: **A chemo-mechano-biological formulation for the effects of biochemical alterations on arterial mechanics: the role of molecular transport and multiscale tissue remodelling.** *J R Soc Interface* 2017, **14**: 20170615.
 53. Keshavarzian M, Meyer CA, Hayenga HN: **Mechanobiological model of arterial growth and remodeling.** *Biomechanics Model Mechanobiol* 2018, **17**:87–101.
- Finite-element and agent-based model for arterial growth and remodeling accounting for cell proliferation, apoptosis, and cell–cell signaling among cells in the three arterial layers, as influenced by mechanics.
54. Loerakker S, Stassen OMJA, Ter Huurne FM, Boareto M, Bouten CVC, Sahlgren CM: **Mechanosensitivity of Jagged–Notch signaling can induce a switch-type behavior in vascular homeostasis.** *Proc Natl Acad Sci U S A* 2018, **115**: E3682–E3691.
 55. Tang Y, Urs S, Boucher J, Bernaiche T, Venkatesh D, Spicer DB, Vary CPH, Liaw L: **Notch and transforming growth factor- β (TGF β) signaling pathways cooperatively regulate vascular smooth muscle cell differentiation.** *J Biol Chem* 2010, **285**: 17556–17563.
 56. Morrow D, Sweeney C, Birney YA, Cummins PM, Walls D, Redmond EM, Cahill PA: **Cyclic strain inhibits Notch receptor signaling in vascular smooth muscle cells in vitro.** *Circ Res* 2005, **96**:567–575.
 57. Zeigler AC, Richardson WJ, Holmes JW, Saucerman JJ: **A computational model of cardiac fibroblast signaling**

- predicts context-dependent drivers of myofibroblast differentiation.** *J Mol Cell Cardiol* 2016, **94**:72–81.
58. Wissing TB, Bonito V, Bouten CVC, Smits AIPM: **Biomaterial-driven *in situ* cardiovascular tissue engineering - a multi-disciplinary perspective.** *npj Regen Med* 2017, **2**:18.
 59. Smits AIPM, Bouten CVC: **Tissue engineering meets immunoengineering: prospective on personalized *in situ* tissue engineering strategies.** *Curr Opin Biomed Eng* 2018, **6**:17–26.
 60. Battiston KG, Ouyang B, Labow RS, Simmons CA, Santerre JP: **Monocyte/macrophage cytokine activity regulates vascular smooth muscle cell function within a degradable polyurethane scaffold.** *Acta Biomater* 2014, **10**:1146–1155.
 61. Battiston KG, Labow RS, Simmons CA, Santerre JP: **Immunomodulatory polymeric scaffold enhances extracellular matrix production in cell co-cultures under dynamic mechanical stimulation.** *Acta Biomater* 2015, **24**:74–86.
 62. Roh JD, Sawh-Martinez R, Brennan MP, Jay SM, Devine L, Rao DA, Yi T, Mirensky TL, Nalbandian A, Udelsman B, *et al.*: **Tissue-engineered vascular grafts transform into mature blood vessels via an inflammation-mediated process of vascular remodeling.** *Proc Natl Acad Sci U S A* 2010, **107**:4669–4674.
 63. Hibino N, Mejias D, Pietris N, Dean E, Yi T, Best C, Shinoka T, Breuer C: **The innate immune system contributes to tissue-engineered vascular graft performance.** *FASEB J* 2015, **29**:2431–2438.
 64. Miller KS, Lee YU, Naito Y, Breuer CK, Humphrey JD: **Computational model of the *in vivo* development of a tissue engineered vein from an implanted polymeric construct.** *J Biomech* 2014, **47**:2080–2087.
 65. Ballotta V, Driessen-Mol A, Bouten CVC, Baaijens FPT: **Strain-dependent modulation of macrophage polarization within scaffolds.** *Biomaterials* 2014, **35**:4919–4928.
 66. Adams S, Wuescher LM, Worth R, Yildirim-Ayan E: **Mechano-immunomodulation: mechanoresponsive changes in macrophage activity and polarization.** *Ann Biomed Eng* 2019. <https://doi.org/10.1007/s10439-019-02302-4>.
 67. Bonito V, De Kort BJ, Bouten CVC, Smits AIPM: **Cyclic strain affects macrophage cytokine secretion and extracellular matrix turnover in electrospun scaffolds.** *Tissue Eng A* 2019, **25**:1310–1325.
 68. Ristori T, Obbink-Huizer C, Oomens CWJ, Baaijens FPT, Loerakker S: **Efficient computational simulation of actin stress fiber remodeling.** *Comput Methods Biomech Biomed Eng* 2016, **19**:1347–1358.
 69. Latorre M, Humphrey JD: **A mechanobiologically equilibrated constrained mixture model for growth and remodeling of soft tissues.** *Z Angew Math Mech* 2018, **98**:2048–2071.
- Steady-state formulation of the constrained mixture growth and remodeling theory that enables the prediction, with a limited computational cost, of the final tissue state resulting from long-term growth and remodeling. This can be adopted to investigate the effects of growth and remodeling laws and parameters on the asymptotic tissue state.
70. Latorre M, Humphrey JD: **Critical roles of time-scales in soft tissue growth and remodeling.** *APL Bioeng* 2018, **2**, 026108.
- Generalization of the steady-state formulation of the constrained mixture growth and remodeling theory for systems reaching an asymptotic quasi-static equilibrium, achieved when the turnover rate of tissue components is much faster than the gradual change of mechanical stimuli.
71. Gasser TC, Grytsan A: **Biomechanical modeling the adaptation of soft biological tissue.** *Curr Opin Biomed Eng* 2017, **1**:71–77.
 72. Szafron JM, Breuer CK, Wang Y, Humphrey JD: **Stress analysis-driven design of bilayered scaffolds for tissue-engineered vascular grafts.** *J Biomech Eng* 2017, **139**:121008.
 73. Szafron JM, Ramachandra AB, Breuer CK, Marsden AL, Humphrey JD: **Optimization of tissue-engineered vascular graft design using computational modeling.** *Tissue Eng C Methods* 2019, **25**:561–570.
- A computational framework that integrates a neovessel development algorithm with a surrogate management framework with the aim to optimize scaffold design parameters such that neovessel remodeling enables the establishment of certain long-term geometrical and functional targets.